Tuning PID Controller Using Artificial Intelligence Techniques

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Abstract: This paper studies PID controller parameters improving using the genetic algorithm, modified genetic algorithm, particle swarm optimization techniques. The proposed techniques compared with conventional PID controllers tuned by Ziegler-Nicholas technique using two applications DC-Motor and AVR system. The overshoots, rise time and settling time with the proposed techniques are better than these of the conventional PID controllers. The effectiveness of the proposed techniques is confirmed using MATLAB M-files software and genetic algorithm toolbox.

Keywords: *PID-controller, DC-Motor, AVR system, Genetic Algorithm, Modified Genetic Algorithm, Particle Swarm Optimization.*

I. Introduction

Proportional-Integral-Derivative (PID) controller is one of the earliest control technique and is still used widely in industrial because of its easy implementation, robust performance and being simple of physical principle of parameters. For achieving appropriate closed loop performance, three parameters of the PID controller must be tuned [1].

Tuning methods of PID parameters are classified as traditional and intelligent methods. Conventional methods such as Ziegler and Nichols method which is hard to determined optimal PID parameters and usually is not caused good tuning, it produces oscillations and big overshoot [2].

Artificial intelligent approaches such as genetic algorithm and particle swarm optimization have been proposed which have received much interest and have been applied successfully to solve the problem of optimization PID controller Parameters.

II. Artificial Intelligence optimization techniques

Genetic Algorithms (GA's) are a stochastic global search method that simulators the process of natural

evolution and they are a family of computational models inspired by evolution. The genetic algorithm starts with no knowledge of the correct solution and depends entirely on responses from its environment and evolution operators (i.e. reproduction, crossover and mutation) to arrive at the best solution. By starting at several independent points and searching in parallel, the algorithm avoids local minima and converging to sub optimal solutions. In this way, GAs have been shown to be capable of locating high performance areas in complex domains without experiencing the difficulties associated with high dimensionality, as may occur with gradient decent techniques or methods that rely on derivative information [3]. Figure (1) shows the steps of creating and implementing the genetic algorithm.

The genetic Algorithm advantages are:

- Optimization with continuous or discrete variables.
- Derivative information isn't required.
- Simultaneously searching from a wide sampling of the cost surface.
- Dealing with a large number of variables.
- Fitting for parallel computers.
- Optimization of the variables with extremely complex cost surfaces (they can jump out of a local minimum).
- Supply a list of optimum variables, not just a single solution.
- Encode the variables so that the optimization is done with the encoded variables.
- Working with numerically generated data, experimental data, or analytical functions.

These advantages are exciting and produce better results when traditional optimization approaches fail.

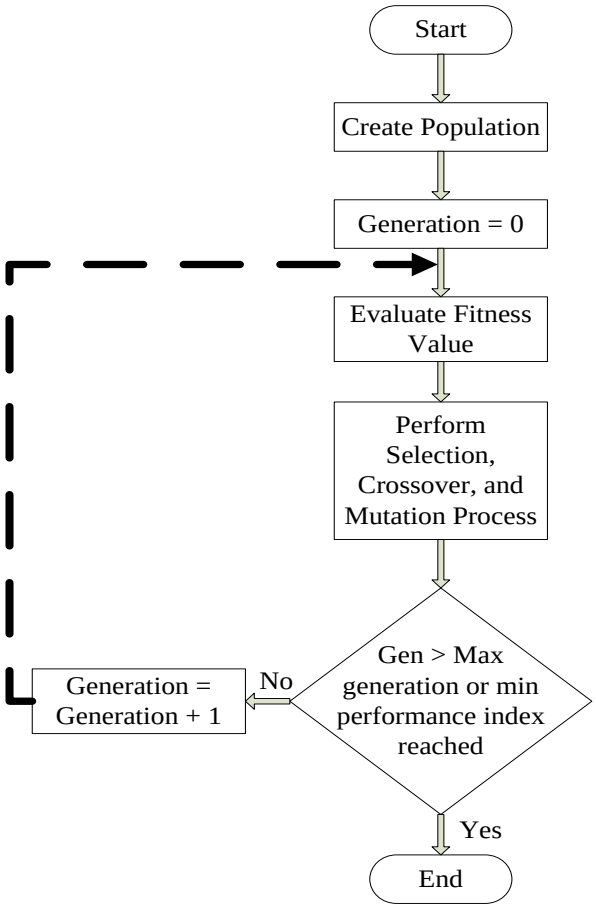

Figure (1): Genetic Algorithm Flowchart

The Particle Swarm Optimization (PSO) was inspired from the computer simulation of the social behavior of bird flocking. Computer graphics simulate

the complicated flocking behavior of birds [4]. In a PSO system, a swarm of individuals (called particles or intelligent agents) fly through the search space. Each particle represents a candidate solution to the optimization problem. The position of a particle is influenced by the best position visited by itself (i.e. its own experience) and the position of the best particle in its entire population. The best position obtained is referred to as the global best particle. The performance of each particle is measured using a fitness function that varies depending on the optimization problem. Modification of the particles position is realized by the position and velocity information according equations (1) and (2) respectively:

$$v_i^{k+1} = v_i^k + c_1\, rand_1 \times (pbest_i - x_i^k) + c_2\, rand_2 \times (gbest - x_i^k) \quad (1)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1} \quad (2)$$

Where:

v_i^k current velocity of particle i at iteration k,

v_i^{k+1} new velocity of particle i next at iteration $k+1$,

c_1 cognitive acceleration constant (self-confidence),

c_2 social acceleration constant (swarm confidence),

x_i^k current position of particle i at iteration k,

x_i^{k+1} new position of particle i at iteration next iteration k+1 ,

$pbest_i$ personal best of particle i ,

$gbest$ Global best of the population.

Figure (2) illustrates the general flowchart for the PSO technique. The sequence can be described as follows:

Step 1: Generation of initial conditions of each particle.

Initial searching points (x_i^0) and the velocities (v_i^0) of each particle are usually generated randomly within the allowable range. The current searching point is set to $pbest$ for each particle. The best evaluated value of $pbest$ is set to $gbest$ and the particle number with the best value is stored.

Step 2: Evaluation of searching point of each particle.

The objective function is calculated for each particle. If the value is better than the current pbest value of the particle, then pbest is replaced by the current value. If the best value of pbest is better than the current gbest, the gbest value is replaced by the best value and the particle number with the best value is stored.

***Step 3*:** Modification of each searching point.

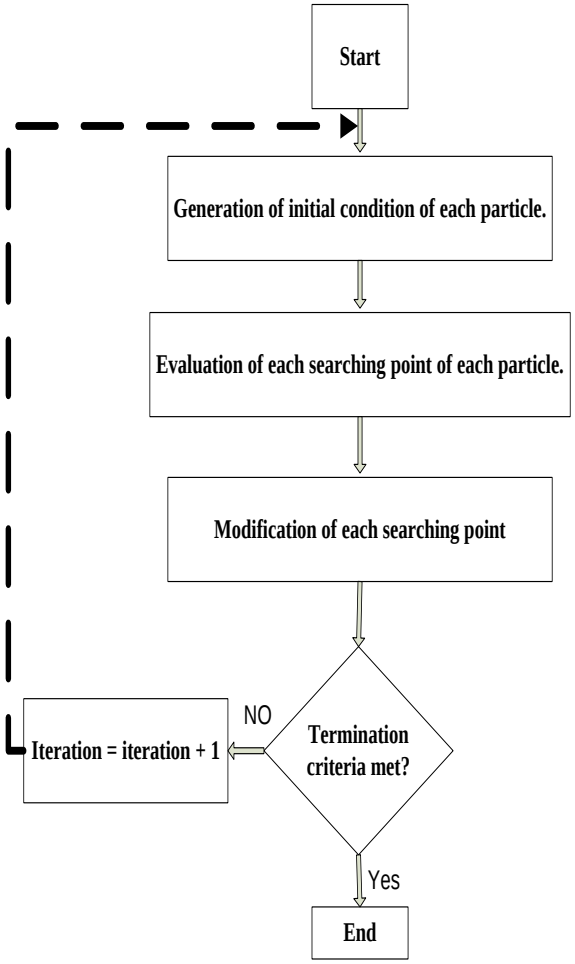

Figure (2): The Particle Swarm Optimization Flowchart [5].

III. GA, MGA and PSO Tuning PID Controller:

Figure (3) illustrates the block diagram of tuning PID-controller using the Genetic Algorithm (GA), the Modified Genetic Algorithm (MGA) and the Particle Swarm Optimization (PSO) applying for the two models DC-Motor and the Automatic Voltage Regulator (AVR) system.

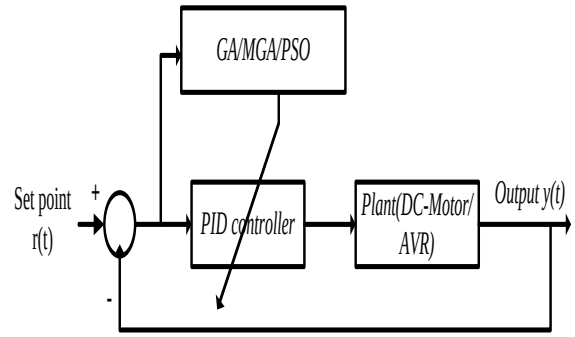

Figure (3): Tuning PID Controller with GA, MGA and PSO.

III.1. PERFORMANCE INDICES:

The performance index (objective function) can be calculated and used to evaluate the system's performance, whether the aim is to improve the design of a system or to design a control system.

a) Integral of the Square of the Error (ISE):

ISE is determined by equation (3); the squared error is mathematically convenient for analytical and computational purposes.

$$ISE = \int_0^T e^2(t)\,dt \qquad (3)$$

b) Integral of the Absolute Magnitude of the Error (IAE):

$$IAE = \int_0^T |e(t)|dt \qquad (4)$$

IAE gets the absolute value of the error to remove negative error components. *IAE* is particularly useful for computer simulations studies, calculated by equation (4) [6].

c) Integral of the Time Absolute Magnitude of the Error (ITAE):

ITAE weights the error with time and hence emphasizes the error values later on in the response rather than the initial large errors. As shown in equation (5).

$$ITAE = \int_0^T t\,|e(t)|dt \qquad (5)$$

In Modified Genetic Algorithm (MGA) the objective function is illustrated as in equation (6) below [7].

$$\boldsymbol{O} = \propto (\boldsymbol{Performance\ Index}) + \boldsymbol{\beta}(\boldsymbol{t_s}) + \boldsymbol{\delta}(\boldsymbol{OS}) \qquad (6)$$

IV. Applications:

These papers apply the proposed tuning methods on two models and compare the results to Z-N tuning method.

IV.1. Position Control of DC-Motor:

This paper use the DC-Motor as the first model with the following specifications:-

2hp,230 v, 8.5 amperes, 1500 rpm

R_a(Armature resistance) = 2.45Ω,

L_a(Armature inductance) = 0.035H,

K_b(Back *EMF*) = 1.2Vs/rad, J_m(Moment of inertia) = 0.022 kgm^2, B_m(Frictional constant) = 0.5*10^{-3}(Nms/rad)

Then the transfer function of DC-Motor is as illustrated in (7) [8].

$$\frac{\theta(s)}{V_a(s)} = \frac{1.2}{0.00077s^3+0.0539s^2+1.441s} \qquad (7)$$

The proposed methods used *MATLAB M-files* for the two models, GA, MGA and PSO. Figures (5) and (6) present respectively the step response of DC-Motor with PI-Controller and PID-Controller with GA, MGA and PSO.

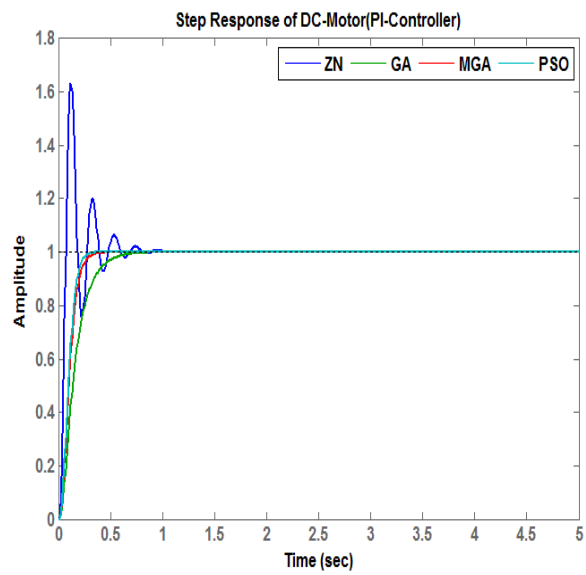

Figure (5): Step Response of DC-Motor with PI-Controller using ZN, GA, MGA and PSO.

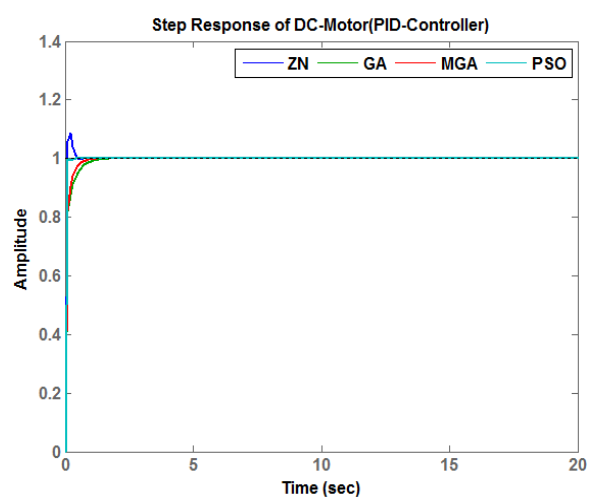

Figure (6): Step Response of DC-Motor with PID-Controller using ZN, GA, MGA and PSO.

IV.2. Automatic Voltage Regulator (AVR) System:

The role of an AVR is to hold the terminal voltage magnitude of a synchronous generator at a specified level. A simple AVR system comprises four main components, namely amplifier, exciter, generator, and sensor [9]-[10]. The reasonable transfer function of these components may be represented, respectively, as shown in (8), (9), (10) and (11) [11].

$$\frac{V_R(s)}{V_e(s)} = \frac{K_A}{1+\tau_A S} \quad ,K_A = 10\ ,\ \tau_A = 0.1\ s \qquad (8)$$

$$\frac{V_F(s)}{V_R(s)} = \frac{K_E}{1+\tau_E S} \quad ,K_E = 1,\ \tau_E = 0.4\ s \qquad (9)$$

$$\frac{V_t(s)}{V_F(s)} = \frac{K_G}{1+\tau_G S} \quad ,K_G = 0.7,\ \tau_G = 1\ s \qquad (10)$$

$$\frac{V_s(s)}{V_t(s)} = \frac{K_R}{1+\tau_R S} \quad ,\ K_R = 1, \tau_R = 0.01\ s \qquad (11)$$

The block diagram of AVR system with PID-controller is presented in figure (4).

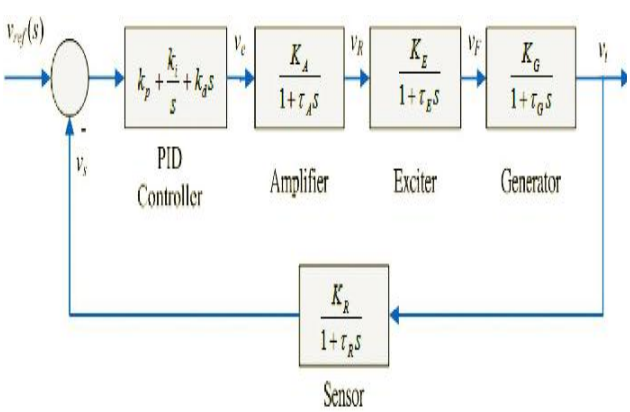

Figure (4): AVR System with PID-Controller [12].

Figures (7) and (8) present respectively the step response of AVR system with PI-Controller and PID-Controller with genetic algorithm modified genetic algorithm and particle swarm optimization.

Tables (1) and (2) content the results for all techniques discussed in this proposal.

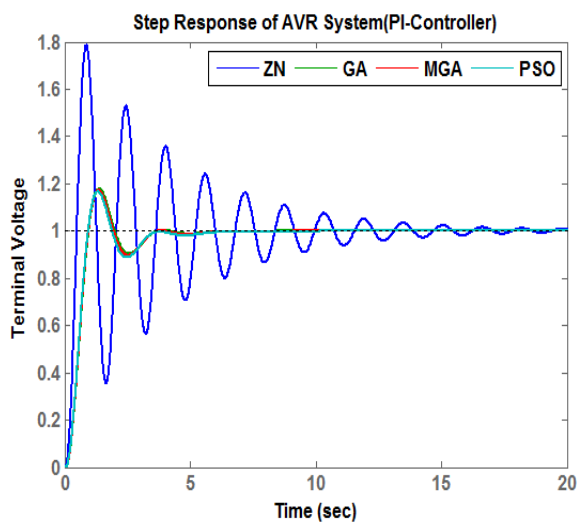

Figure (7): Step Response of AVR System with PI-Controller using ZN, GA, MGA and PSO.

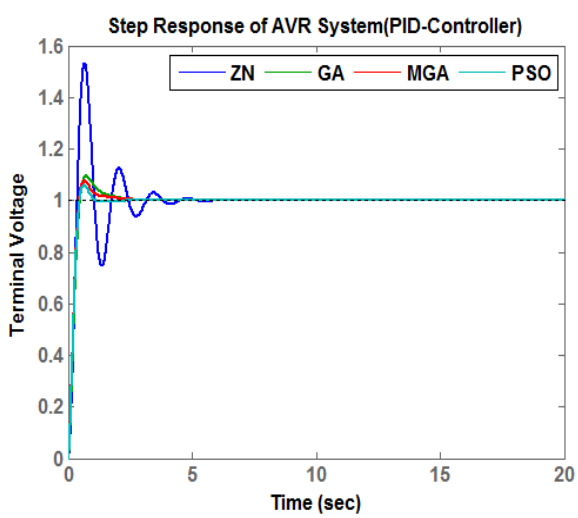

Figure (8): Step Response of AVR System with PID-Controller using ZN, GA, MGA and PSO.

Table (1): Comparison of Results for DC-Motor.

Tuning Method	DC-Motor with PI-Controller					
	K_p	K_i	K_d	Max. OS	T_r(Sec)	T_s(Sec)
ZN	37.8	151.2	-	63	0.0423s	0.752s
GA	6.94	0.011	-	0.0273	0.291s	0.547s
MGA	10.589	0.012	-	0.0128	0.157s	0.288s
PSO	11.4274	0.002	-	0.00184	0.14s	0.246s
DC-Motor with PID-Controller						
ZN	49.41	329.4	1.852	16.9	0.0298s	0.355s
GA	14.859	0.01	3.589	0.00542	0.0258s	0.771s
MGA	19.873	0.009	3.221	0.00273	0.184s	0.541s
PSO	32.6051	0.0125	1.0441	0.00141	0.08s	0.0981s

Table (2): Comparison of Results for AVR System.

Tuning Method	AVR System with PI-Controller					
	K_p	K_i	K_d	Max. OS	T_r(Sec)	T_s(Sec)
ZN	1.093	1.203	-	78.7	0.289s	15.2s
GA	0.413	0.251	-	18	0.604s	3.38
MGA	0.424	0.243	-	17.5	0.598s	3.36s
PSO	0.4379	0.2329	-	16.9	0.59s	3.35s
AVR System with PID-Controller						
ZN	1.457	2.006	0.149	53.4	0.249s	3.63s
GA	0.893	0.848	0.236	9.37	0.326s	1.64s
MGA	0.96	0.826	0.265	7.82	0.301	1.22
PSO	0.9564	0.6725	0.2613	5.95	0.301	0.926

V. Conclusion:

The PID controller based on Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) has better steady state response and performance indices than conventional methods and it took only less than the second to solve the problem. The Particle Swarm Optimization (PSO) algorithms is the best controller which presented satisfactory performances and good Robustness (no overshoot, minimal rise time, steady state error is 0).

VI. References:

[1] K.J.Astrom, T. Hagglund, “PID Controllers: Theory, Design and Tuning”, International Society for Measurement and Con., 2nd Ed., pp. 200-210, 1995.

[2] F.D.Bianchi, R.J. Mantz, C.F. Christiansen, “Multivariable PID

Control with Set-Point Weighting via BMI Optimization", Automatica, Vol. 44, pp. 472-478, 2008.

[3] Ali Zilouchian and Mohammad Jamshidi, "Intelligent control systems using soft computing methodologies", by CRC Press LLC, 2001.

[4] NELENDRAN PILLAY, "A PARTICLE SWARM OPTIMIZATION APPROACH FOR TUNING OF SISO PID CONTROL LOOPS", DURBAN UNIVERSITY OF TECHNOLOGY, 2008.

[5] Ali Marzoughi, Hazlina Selamat, Mohd Fua'ad Rahmat and Herlina Abdul Rahim, " Optimized proportional integral derivative (PID) controller for the exhaust temperature control of a gas turbine system using particle swarm optimization", International Journal of the Physical Sciences Vol. 7(5), pp. 720-729, 2012.

[6] M. A. Tammam, M. A. S. Aboelela, M. A. Moustafa, A. E. A. Seif; " Fuzzy Like PID Controller Tuning by Multi-Objective Genetic Algorithm for Load Frequency Control in Nonliear Electric Power Systems", International Journal of Advances in Engineering & Technology, , Vol. 5, Issue 1, Nov. 2012, pp. 572-583.

[7] AYTEKIN BAGIS, "Determination of the PID Controller Parameters by Modified Genetic Algorithm for Improved Performance", JOURNAL OF INFORMATION SCIENCE AND ENGINEERING 23, 1469-1480 (2007).

[8] Neenu Thomas, Dr. P. Poongodi, "Position Control of DC Motor Using Genetic Algorithm Based PID Controller", Proceedings of the World Congress on Engineering 2009 Vol II, London, U.K., 2009.

[9] Zwe-Lee Gaing, "A Particle Swarm Optimization Approach for Optimum Design of PID Controller in AVR System", IEEE TRANSACTIONS ON ENERGY CONVERSION, VOL. 19, NO. 2, JUNE 2004.

[10] Ching-Chang Wong, Shih-An Li and Hou-Yi Wang, "Optimal PID Controller Design for AVR System", Tamkang Journal of Science and Engineering, Vol. 12, No. 3, pp. 259_270 (2009).

[11] S. F. M Kheder, M. E. Ammar, M. A. Moustafa Hassan;"Multi objective genetic algorithm controller's Tuning for non-linear automatic voltage regulator" Accepted for presentation at CoDIT 2013, Tunis, May 6-8, 2013

[12] Ching-Chang Wong, Shih-An Li and Hou-Yi Wang, "Optimal PID Controller Design for AVR System", Tamkang Journal of Science and Engineering, Vol. 12, No. 3, pp. 259_270 (2009).